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PI Name:	Mitchell, Cary Ph.D.		
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PI Email:	cmitchel@purdue.edu	**Fax:**	FY 765-494-0391
PI Organization Type:	UNIVERSITY	**Phone:**	765-494-1347
Organization Name:	Purdue University		
PI Address 1:	Department of Horticulture and Landscape Architecture		
PI Address 2:	625 Agriculture Mall Dr		
PI Web Page:	http://www.alsnscort.org		
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Task Description:

Providing clean water and air for long-term exploration missions is a top priority for the NSCORT research team. The Air and Water Research Group is focusing on development of innovative technology for wastewater treatment and recycling, drinking water treatment and safety, and removal of trace contaminants from air.
The air/water and solids focus groups of the NSCORT were merged into a single resource-recovery (R2) focus area during year 3 of Center operations, because many R2 projects are now coming together across disciplines. Two research projects came to a close during the past year, a new project started up, and a center-wide analytical effort was initiated within the R2 group. In addition, sensitivity has spread across the Center regarding the need for all research projects to identify steps or points in bioprocessing where hazards to human crews might occur. Potential hazards resulting from biological approaches to bioprocessing include microbial contamination, toxic metabolites in waste feedstocks and/or bioreactor products, and noxious gases released from metabolic or physico-chemical degradation of organic substances in air, water, or solid phases. Absorption, inhalation, or ingestion of hazardous organisms or substances by crews are of particular concern in closed systems.

A new NSCORT approach to degrading crop and food waste (Plant-Based Anaerobic-Aerobic Bioreactor-Linked Operation, or PAABLO) biologically pretreats grey water, reduces solid waste loads, and produces methane for subsequent energy reclamation, without producing copious waste heat. An analogous bio-reactor (Waste-AABLO, or WAABLO) reduces human solid waste in a separate process stream. PAABLO recalcitrant residues are directed to a fungal composter for partial hydrolysis of ligno-cellulose, the hydrolysates are returned to the anaerobic reactor to maximize methanogenesis, energy reclamation, and solids reduction, and the pre-treated grey water is directed to BREATHe I for aerobic co-treatment of air and water. BREATHe I also can remove NH_4^+ from aquarium water and urea from urine in separate process streams. Micro-, nano-, and RO-membranes filter out trace organics and desalt inorganics from BREATHe effluents, and iodine and UVc provide final polishing and fail-safe quality control. Noxious gases emitted from bioreactors and incinerators are processed by BREATHe II biofilters and/or concentrated onto zeolite adsorbent particles. Nitrogen and sulfur-rich adsorbents or eluates are added to fungal composters and microbial bioreactors to stimulate the growth of reactor organisms feeding on N- and S-deficient ligno-cellulose and enhance their hydrolysis of feedstock polymers. Consumption by Tilapia fish of microbial biomass containing amino acids essential for fish development enables those fish to indirectly reduce the load of recalcitrant crop waste while supplementing crew diets with small amounts of animal protein.

By using products of one waste treatment as substrates for another, by focusing on bioregenerative approaches to primary processing and limiting physico-chemical approaches to secondary processing and polishing functions, and by analyzing for bio-hazards at critical process points, the NSCORT R2 group is significantly reducing ESM for resource recovery while minimizing potential risks in closed systems.

Cary Mitchell for Al Heber, R2 Focus Area Lead For BREATHe: 1 - Construct and operate a representative bench-scale biotrickling filter treatment system. 2 - Optimize key design parameters identified through a first generation model, specifically packing material surface area and liquid velocity. 3 - Assess biodegradability of surfactants of concern, sodium laureth sulfate (SLES) and disodium cocoamphodiacetate (DSCADA). 4 - Screen for potential surfactant degradation byproducts in bioprocessor effluent. 5 - Determine Monod kinetic parameters for surfactants of interest. 6 - Quantify Henry's constants for target gas contaminants (NH_3 & H_2S) when present as gas mixtures and in the presence of surfactant.

For Membrane Filtration Processes: 1 - Modify commercially available NF membranes to improve efficiency for targeted contaminant removal 2 - Synthesize novel new NF membranes with increased flux and rejection properties 3 - Modify commercially available MF membranes to mitigate biofouling

For Iodine Chemical Water Disinfection: 1 - Determine whether complementary disinfection using UV radiation and iodine will provide more effective microbial inactivation than either method alone. 2 - Develop a chemical actinometer that can monitor the UV disinfection process, in addition to providing sufficient quantities of iodine to achieve residual disinfection. The dual purpose of the chemical actinometer may reduce chemical restock requirements. 3 - Investigate whether the microbial check valve (MCV), which was developed in the 1970s by NASA, can be used to store and deliver the photochemically-produced iodine. 4 - Evaluate the effectiveness of using Vitamin C to remove iodine from potable water, while providing a beneficial residual for the astronauts.

Water Disinfection UV Irradiation (Physical): 1 - Develop a numerical method for evaluation of process performance of various hypothetical reactor geometries and operational parameters 2 - Validate the modeling tool by conducting microbial inactivation experiments with existing hardware and simulating the process numerically 3 - Identify and procure a non-mercury-containing UV source suited for long-term space missions 4 - Perform UV dose-response experiments with challenge microorganism and non-mercury-containing UV source 5 - Devise and program a model for solving the intensity field around the chosen UV source 6 - Use the modeling tool to investigate candidate reactors 7 - Choose the best reactor design based on model results and build and test a physical prototype reactor

For Gas Rivitalization using Biofiltration: 1 - Optimize the design and operation of ALS biofiltration process to maximize the trace contaminant removal efficiency, from perspectives of both microbiology and engineering, while minimizing the Equivalent System Mass (ESM). 2 - Develop macrokinetics and microkinetics models for ALS biological gas phase treatment systems. 3 - Delineate technological barriers for use of biofiltration in ALS systems.

For LiFT: 1 - Theoretical modeling of freeze concentration process and contaminant migration 2 - Detailed evaluation of inorganic and organic migration to product water 3 - Detailed evaluation of possible low level contaminants including biological and antibiotic / pharmaceutical compounds entering product stream 4 - Detailed evaluation of potential urea recovery from brine solution

Rationale for HRP Directed Research:

	Development of new water and air-treatment technologies for environmental protection Will impact air-treatment technology and air quality control strategies used in residential and commercial buildings and in commercial aircraft.
Research Impact/Earth Benefits:	Further development of fundamental photochemical reactor theory as it relates to systems used for disinfection of water and air.

For BREATHe I:
Operation of Bench Scale Biotrickling Filters. Results from the first generation model presented in last year's report indicated that contaminant removal rates in the liquid phase are highly dependent on packing material surface area and liquid velocity with removal rates increasing with increases in the former and decreasing with increases in the latter. Based on these results, it was determined that packing materials are an extremely important design consideration since not only does their surface area have an effect on microbial activity, but the geometry and size can have a significant impact on liquid distribution and velocity though a biotrickling filter. The effects of several packing materials on liquid velocity, or residence time, in the trickling filters were quantified through a series of tracer tests (data not shown). Tri-pack packing material was used for bench-scale experiments conducted in 2004 where steady state TOC removal was on average 62%. It was hypothesized that use of packing material with a higher surface area would increase process performance due to a combination of increased area for microbial growth and decreased liquid velocity. In all cases, the reactor startup phase took approximately 40 days during which point TOC removal was highly variable. A one sided t-test showed that removal rates were not significantly improved using Biobale or Bee-cell packing material at the 95% significance level. At this point it was hypothesized that TOC removal may be limited by liquid distribution in the reactor resulting in incomplete utilization of packing material surface area. The recommended hydraulic loading rate to optimize liquid distribution in trickling filters is 1.94 m/hr, which would require a recirculation rate of 75X for the constructed bench-scale reactors. A recirculation rate of 20X was employed for data presented thus far. On day 107 of operation, the recirculation rate was increased from 20X to 75X . Again, TOC removal was not improved. During these experiments, surfactants were analyzed by LC-MS methods to further characterize the composition of effluent TOC. Interestingly, surfactant removal was on average 99%. It has come to our attention that biodegradation of SLES and/or DSCADA may result in a recalcitrant byproduct contributing to effluent TOC. Biodegradability of SLES and DSCADA. A set of shaker flask experiments was conducted to assess the biodegradability of SLES and DSCADA. Prior to the start of experiments, a mixed culture of bacteria was acclimated to SLES and DSCADA degradation. Three replicate flasks were prepared that contained 248 ppm SLES and 5% by volume SLES acclimated bacteria and another three flasks were prepared that contained 218 ppm DSCADA and 5% by volume DSCADA acclimated bacteria. One experiment set was conducted using BREATHe graywater simulant as a nutrient medium and another was tested using a minimal salts medium (MSM). SLES was readily degraded in both nutrient media with TOC approaching zero as the tests ended . Conversely, TOC levels did not reduce below 40 mg/L in flasks containing DSCADA. Samples were also tested for COD, with similar trends observed . COD approached zero in flasks containing SLES and did not go below 100 mg/L in flasks containing DSCADA. DSCADA degradation also was examined in a rich nutrient medium, tripticase soy broth, so that a more diverse microbial population could develop that may support complete degradation of DSCADA. In this situation, TOC did not go below 200 mg/L, indicating that even easily degradable carbon present in tripiticase soy broth was not degraded. It is likely that a DSCADA degradation byproduct is toxic to bacteria inhibiting their ability to grow and metabolize organic carbon. Degradation pathways for amphoteric surfactants, the group to which DSCADA belongs, have not been well characterized so it is difficult to predict the byproduct that creates toxicity issues. Toxicity issues have previously been noted for amphoteric surfactants as the quaternary nitrogen group existing in alkylamphoacetate surfactants has been found to cause antimicrobial activity (Domsch, 1995). However, this was typically overcome by bacterial acclimation to the surfactant of interest, which did not work in the case presented. Toxicity issues related to DSCADA degradation may be the cause of limited TOC removal in the BREATHe I system. Screening forPotential Surfactant Degradation Byproducts. The buildup of any contaminant within an advanced life support system could pose serious health risks to crew members. Therefore, it is important to identify surfactant degradation pathways and screen for potential byproducts in bioprocessor effluent. BDoxy biosensors were used to confirm degradation byproducts for Polyalcohol Ethoxylate (PAE), which is commonly used in laundry and dish wash detergents. BDoxy biosensors are 96 well plates containing an oxygen sensitive fluororphore. Substrate and bacteria can be added to each well and resulting fluorescence is indicative of microbial activity. Several substrates that have been identified in the literature as potential degradation byproducts of PAE were added to wells containing bacteria acclimated to PAE degradation. Substrates studied included ethylene glycol (EG), diethylene glycol, triethylene glycol, hexaethylene glycol, octaethylene glycol, glycolic acid, formic acid, glyoxilic acid, and oxalic acid. A positive response significantly higher than a control population would indicate acclimation of bacteria to that substrate and thus confirm presence as a degradation byproduct of the parent compound. Fluorescence of EG compounds were normalized relative fluorescence units (NRFU). All of the compounds had a significantly higher response than the control population except for octaethylene glycol. The only organic acid that showed a response was glycolic acid. Polyethylene glycols are clearly important degradation byproducts for PAE. It is quite possible that EG could be present in effluent of a bioprocessor treating PAE, particularly because it's degradation is the last step in the multi step process. EG is highly toxic for human consumption with effects such as vomiting, drowsiness, coma, respiratory failure, gastrointestinal upset, cardiopulmonary effects, renal damage, and potential fatality. EG is also a predicted degradation byproduct of SLES degradation. BREATHe effluent will be screened for EG in the future using a UV method that has previously been developed. If present in bioprocessor effluent, the fate of EG through post treatment processes such as membranes should be assessed. Biokinetics of Sufactant Degradation. Monod kinetics have been determined for surfactants of interest SLES, DSCADA, and PAE using respirometry methods developed by Grady et al., 1989. Parameter estimates represent averages of six replicate tests. Mass Transfer of Gas Constituents. Henry's constants were determined for each of the combination of gases in the presence and absence of surfactants. Initially, 600 mL of air and 400 mL of solvent were allowed to equilibrate and gas measurements were made. Known volumes of calibration gas were injected into the system and the system was again allowed to equilibrate. Note that only small volumes of gas were injected into the system as Henry's Law is only valid for dilute solutions. Distribution curves allowed the Henry's constants to be determined for combinations of H_2S, CO_2, and NH_3 gases. Surfactants did not have a significant effect on solubility for any of the gas combinations tested. The largest effect on abiotic system mass transfer seemed to be the presence of CO_2, as the solubility of both NH_3 and H_2S seemed to slightly increase. Due to the large range of Henry's constant in the literature, none of the gas mixtures and/or surfactants can be attributed to changes in solubility.

For Membrane Processes in ALS:

NF (Nano-Filtration) membranes are being synthesized at Howard University to decrease fouling and increase salt rejection for water reuse applications. NF membranes operate under much lower operating pressure than RO (Reverse Osmosis) membranes, resulting in significantly reduced ESM for the system. Thus, focus is presently on developing new NF membranes to be used as alternatives to traditional high pressure RO membranes for recovery of space mission wastewater. Thrust 1: Synthesis of novel new NF membranes with increased flux and rejection properties Nanofiltration membranes modified to increase negative surface charge exhibited higher salt rejection when compared to unmodified membranes. These membranes were modified by an ion implantation technique, which implanted the electronegative F- ion into the active layer of the membrane in order to render the membrane more negative and increase rejection by electrostatic repulsion. Based on these studies, membrane synthesis experiments were undertaken to develop novel new membranes with controllable pore size and surface charge, which will allow for higher flux and rejection. Synthesis and Characterization. Cellulose acetate (CA) membranes were spun onto a polypropylene backing. The rejection properties of the CA membranes were varied by changing the initial concentration and spinning quenching conditions. These experiments were undertaken as part of a collaboration with the Nanobiotechnology Center at Cornell University. Effective pore sizes were measured by pressure driven rejection of uncharged solutes. The limiting rejection of an uncharged solute is directly proportional to the ratio of solute radius to pore radius. The surface charge was evaluated by electrokinetic analysis/streaming potential measurements. The zeta potential of the membrane, determined from electrokinetic analysis (EKA) was measured as a function of pH. Results. The zeta potential (and thus the charge of the CA membranes) is negative and increases with increasing pH. The isoelectric point is pH 3.5, but the maximum zeta potential for basic pH increases with acetate concentration. Uncharged solute rejection shows a 1.7 nm pore radius for 25% acetate membranes, classifying them as nanofiltration with filtration mechanisms dependent on charge and steric hindrance. These properties are ideal for separation of salts such as NH4 +, Na+, and Cl-. Membranes with molecular weight cutoffs (MWCOs) of 300, 350, and 600 Da have been fabricated, which should also reject residual surfactant monomers that may remain following BREATHe. Experiments are ongoing to determine the rejection of salts, organics and surfactant monomers from the synthesized membrane. Thrust 2: Reduction of Biofouling in Microfiltration Membranes using Membrane Modification MF (Microfiltration) membranes will be used as pretreatment for the NF membranes. The large pore size in MF membranes results in lower pressure and higher flux than for the other pressure driven membranes, making MF membranes ideal for the low ESM requirements in ALS NSCORT. The largest problem with the MF membranes will likely to due to biofouling, as these membrane follow directly behind BREATHe, which may leak biosolids. The MF membranes will reject biosolids, but these materials may accumulate on the membrane surface and reduce flux. Experiments were begun to modify MF membranes by grafting acrylic acid onto the surface of these membranes, which will increase hydrophilicity of the surface and smooth the morphology, which reduces biofilm accumulation on the surface. Flux. Polyethersulfone (PES) membranes of pore sizes 0.1 microns and 500 kDa were modified. Filtration experiments were carried out in an Amicon stirred cell at a fixed pressure of 20 psi. Flux experiments were conducted with clean water to establish initial permeability, then simulated wastewater was used in order to quantify the biofouling extent of the membranes. Biofouling. Simulated wastewater was filtered for at least 6 hours or until the flux was at steady state. The fouled membrane was rinsed and replaced in the filtration cell and filtered with clean water. The membranes were the placed in a sodium pyrophosphate solution to remove the bacteria. The bacteria were then quantified using the pour plate method. Modification. Acrylic acid was the monomer chosen mainly because of its antimicrobial properties, its ability to reduce irreversible fouling and its hydrophilicity. The membranes were coated with a photoinitiator by soaking them in 0.25M benzophenon in ethanol. The membranes were dried and acrylic acid with the inhibitor removed was placed on the active side of the membrane. The membranes were then UV irradiated and after polymerization the membranes were extracted with methanol/water solution and washed with sodium hydroxide Ultrapure water to remove any non-grafted polymer, monomer or residual initiator. Results: Flux declines of the modified and unmodified MF membranes are shown in using clean water. The modified membrane showed less severe flux decline than the unmodified membrane, most likely due to the increased hydrophilicity of the modified membrane. Normalized flux values were obtained using synthetic wastewater. Wastewater is a surrogate for BREATHe effluent, with contaminant values reported from Purdue University. The modified MF membrane exhibited enhanced flux after modification, while the modified UF membrane showed reduced flux following modification. Mechanisms governing this behavior are being investigated.

For Iodine Chemical Water Disinfection:

Complementary Disinfection: Experiments were performed to investigate the independent and combined effects of UV and iodine using Bacillus subtilis spores as the challenge microorganism. Spore forming bacteria have been recognized as one of the “hardiest” forms of life on Earth. B. subtilis have been widely studied as a surrogate for other spore-formers in laboratory settings because they are non-pathogenic and their resistance to many forms of disinfection has been documented in published literature. Furthermore, they have been identified by the United States Environmental Protection Agency (USEPA) as a challenge microorganism suitable for the design of UV disinfection systems. The results were collected for B. subtilis spore inactivation by UV irradiation at 222 nm, 254 nm and 282 nm. For each of the wavelengths investigated, the reduction in the concentration of viable spores occurred in three distinct stages: a lag period (or shoulder) where disinfection is slow, until a “threshold dose” is reached, at which a straight-line (first-order) decrease in the viable population is observed. The first-order portion of the curve is followed by a transition period where the disinfection effectiveness is slowed until finally, a tailing period is observed. To investigate B. subtilis spore susceptibility to iodine, batch experiments were conducted. To examine the validity of the CT concept, two initial iodine concentrations were used. The results suggest that the inactivation behavior of the spores can be estimated by a first order process (i.e. no shoulder or tailing regions were observed). Based on this model, a dose of approximately 1400 mg-min/L is required to achieve 2-log10 units of inactivation. Sequential disinfection experiments were performed by irradiating a suspension of B. subtilis spores using a collimated UV source, and then subjecting it to iodination in a batch system. For each wavelength investigated (254 nm and 282 nm) the spores were exposed to three different UV doses. The first dose corresponded to the shoulder region of the UV dose-response curve, the second dose corresponded to the region of first-order decay, and the third dose corresponded to the region just before tailing. Following each of these doses, the spores were exposed to a known concentration of iodine and the dose-response behavior was observed. The effects of sequential disinfection are most apparent when the spores were pre-irradiated (254 nm) with a dose of 17.7 mJ/cm2. This dose corresponds to the portion of the UV Dose-response curve in between the shoulder and tailing regions. The doses of 4.5 mJ/cm2 and 36.5 mJ/cm2 represent the shoulder and tailing portions of the UV dose-response curve, respectively. Neither of these doses resulted in increased inactivation by iodination. At a dose of 4.5 mJ/cm2, it is hypothesized the spores may not have acquired enough damage to increase their susceptibility to iodination. At a dose 36.5 mJ/cm2, the majority of the spores had already been inactivated (over four-logs). It is hypothesized that some spores within the population display inherent resistance to UV irradiation; it is possible that these attributes of the spore

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also result in increased resistance to chemical disinfectants, such as iodine. The results of the effects of sequential disinfection when the spores were pre-irradiated using a UV source that emitted at 282 nm indicate that only small increases in iodine susceptibility were observed for the UV doses that correspond to the shoulder and near-tailing regions (4.5 mJ/cm2 and 40.8 mJ/cm2, respectively). However, when the spores were pre-irradiated with a UV dose that corresponds to the first-order portion of the UV dose-response curve (22.5 mJ/cm2), iodination resulted in over 2-logs more inactivation than was observed when the spores were not pre-irradiated. It should also be noted that for pre-irradiation doses of 22.5 and 40.8 mJ/cm2, tailing behavior was observed. Statistical analysis of the data indicates that differences between inactivation kinetics for sequential disinfection compared to iodination alone were statistically significant (p=0.05) for all of the 282 nm pre-irradiation experiments. For the 254 nm experiments, differences in inactivation kinetics were only statistically significant (p=0.05) for the 17.7 mJ/cm2 pre-irradiation experiment. A numerical model was developed to describe the effects of UV and iodine on B. subtilis spores. The model was based on the hypothesis that phenotypic variation in a microbial population is responsible for the existence of a small fraction of the microbial population that has physiological characteristics that enhance its chances of surviving various forms of externally-applied stress, despite the susceptibility of the population majority. Chemical Actinometry: Iodide (I-) and iodate (IO3-) are used as the chemical actinometer to monitor the efficacy of the UV system. The chemical actinometer system to be employed in this application consists of a small capillary tube installed within the UV reactor through which an I-/IO3- solution is pumped. When exposed to UV radiation, this solution will form triiodide (I3-); Laboratory experiments have been performed with collimated-beam devices using conventional low-pressure Hg lamps, as well as excimer sources to determine the quantum yield for photochemical production of triiodide from the iodide/iodate actinometer at UV wavelengths of 222 nm, 254 nm, and 282 nm. Additional experiments have been performed using a small quartz capillary tube installed parallel to the lamp wall. The iodide-iodate actinometer solution was pumped through the capillary tube at a constant flow rate. The data from these experiments confirm the application of the iodide-iodate actinometer as a means to measure the local irradiance emitted from an excimer source. The capillary tube was located at various distances from the lamp surface while the iodide-iodate actinometer was pumped through the capillary at a constant flowrate. By positioning the capillary at a location within the proposed reactor where minimum dose delivery is expected, proper operation of the UV disinfection system can be monitored and verified. For instance, at a given distance from the lamp, the actinometer solution should produce a given amount of triiodide based on the calculated exposure time. Correlating this data to a standard curve, will allow the UV system to be monitored. Residual Disinfectant Production: To limit the use of additional chemicals, the residual disinfectant selected for the closed-loop disinfection system was I2. I2 can be produced using the photo-reacted actinometer solution (containing a mixture of I-, IO3-, I3-, and trace quantities of several other iodine forms) and ascorbic acid (Vitamin C). Vitamin C was selected as a reagent for iodine production because it provides a beneficial anti-oxidant residual for the astronauts. Equation 1 shows the overall chemical reaction that takes place to form I2. As shown in this equation, iodate is the form of iodine that will react with ascorbic acid to form I2. Iodate (IO3-) is present in the photo-reacted actinometer solution at a relatively high concentration (0.1 M).

Equation 1: 2IO3- + 5AA ? 5DHAA + 6H2O + I2

Where: AA = Ascorbic Acid (C6H8O6) DHAA = Dehydroascorbic Acid (C6H6O6)

Following the production of iodine from the reaction of ascorbic acid and iodate, I2 will be added to water that has already been treated by ultraviolet irradiation. The water containing I2 will be stored until ready for consumption. At the point of use, prior to consumption, the I2 and other iodine species will be removed because of potential adverse health effects associated with iodine consumption. Iodine removal will be achieved using ascorbic acid and ion exchange. Ascorbic acid will be added in slight excess to ensure the formation of iodide (I-). Equations 2 through 4 show the reaction of various iodine species with ascorbic acid to form iodide.

Equation 2: I2 + AA+ H2O ? DHAA + 2I- + 2H+ Equation 3: I3- + AA+ H2O ? DHAA + 3I-+ 2H+ Equation 4: 2IO3- + 6AA ? 6DHAA + 5H2O + 2I-

Once all forms of iodine have been converted to iodide, the water will pass through an ion exchange column, where the iodide will be removed. Experiments conducted using Rohm and Haas Amberlite IRA-400Cl ion exchange resin confirm that iodide can be removed from solution via ion exchange and that the process can be described using a Langmuir Isotherm.

For Water Disinfection UV Irradiation (Physical):

During the third year of the NSCORT ALS project, research has been focused on testing the effectiveness of candidate UV sources and utilizing the developed numerical modeling tool to finalize designs for the disinfection reactor that will satisfy constraints of microbial inactivation and ESM minimization. Testing of candidate UV sources. One of the main constraints imposed on the design of UV disinfection reactors for extended space missions is the restriction of use of mercury-containing lamps, as defined by the existing safety regulation documents for the International Space Station (NASA JSC, 2004). Therefore, two alternative non-mercury-containing excimer lamps have been examined as potential UV sources in disinfection reactors. Both excimer lamp systems satisfy the required criteria that they emit radiation of germicidal wavelengths and are efficient enough to be used for water treatment systems. These are the KrCl* (the asterisk notation denotes an excited molecular complex, which has no stable ground state under normal conditions) and the XeBr* excimer lamps, which emit nearly monochromatic radiation at 222 nm and 282 nm, respectively. Experiments were conducted employing the Bacillus subtilis spores as a challenge organism to asses the effect of UV irradiation at wavelengths of 222 nm and 282 nm on spores. Bacillus subtilis were selected for these experiments because they are non-pathogenic to humans and they exhibit relatively high resistance to UV radiation. It can be hypothesized that a system that inactivates these spores effectively will perform well in response to organisms with lower resistance to UV irradiation; most protozoan parasites and vegetative bacteria are less-resistant to UV radiation than B. subtilis spores. Both excimer sources were effective for inactivation of the challenge organism. However, employing radiation at 222 nm was shown to be impractical as the transmission of UV radiation through a representative water sample at 222 nm is much lower that at higher wavelengths of interest, i.e. at 282 nm. This was determined by analyzing effluent samples from the Bio-Regenerative Environmental Air Treatment for Health (BREATHe I) effluent. In the proposed NSCORT ALS water recycling and treatment loop, biological treatment in the BREATHe I reactor is followed by a micro-filtration membrane and UV disinfection. To obtain a representative influent water sample for the UV reactor, effluent samples were collected on two different days from six operating pilot scale BREATHe I reactors, and these samples were then pre-filtered with a 0.45 micrometer membranes to simulate the effect of the micro-filtration membrane. The water samples were then used for the examination of their optical characteristics at 222 nm and 282 nm. The average transmittance values at 222 nm and 282 nm were 16.7 % and 89.9 %, respectively. Therefore, it was

concluded that XeBr* excimer radiation at 282 nm penetrates the aqueous media much more effectively. This characteristic, along with higher degree of electrical efficiency and longer lifetimes for the XeBr* lamp as compared to the KrCl* lamp, influenced the choice of the XeBr* source over the KrCl* source.

UV Intensity field modeling for the XeBr* excimer source. As the physics of excimer formation and UV radiation emission differ from those of UV radiation emitted by mercury-based sources, the Line Source Integration model used to describe the intensity profile emanating from a low-pressure mercury lamp source could not be used for the XeBr* source. The new model for excimer lamps, named the Surface Power Apportionment for Cylindrical Entities (SPACE) model, calculates radiation received at any point around the cylindrical lamp as a contribution from a large number of point sources radiating from the interior annual space of the excimer lamp. The SPACE model also accounts for the refraction, reflection and absorbance effects of the quartz lamp envelope and the media surrounding the lamp. Investigating the hydraulic dynamics and inactivation efficiency of candidate reactors. Computational modeling softwares GAMBIT and FLUENT were used to draw the reactor geometry and solve the fluid flow and microbial trajectories within the reactor for defined operating parameters. The microbial trajectory data was integrated with the SPACE intensity model and the measured inactivation kinetics of Bacillus subtilis spores. This enabled the calculation of overall process efficiency defined as the degree of microbial inactivation achieved by the reactor. Different reactor geometries were compared based on this parameter and their mass, volume and power usage, as these parameters influence the ESM value of the system. The investigated reactor variations included a 90 degree inlet elbow and a straight inlet, three different reactor diameters and internal baffle structures, two different reactor lengths and a range of output power values for a XeBr* excimer lamp with a 2.54 cm diameter. Two investigations were completed regarding reactor geometries with different inlet configurations and baffle design. All reactors exhibited a higher degree of microbial inactivation efficiency with internal baffles as compared to a reactor without any baffles, and the straight inlet configuration provided the most optimal hydraulic conditions. The interior baffle structure around the lamp polarizes the flow and effectively creates a spiral flow pattern through the reactor, influencing the delivery of a more uniform dose distribution UV dose to the microbes traversing the reactor. This is the chosen design. The XeBr* excimer lamp output power specifications have also been determined and a prototype lamp will be custom made for the NSCORT ALS UV disinfection reactor by the manufacturer.

For Gas Rivitalization using Biofiltration:

Simulation of Biofiltration and Indoor Air Quality in the ALS System. A simulation of biofiltration processes was developed to predict the removal efficiency of the gas phase contaminants in the ALS cabin during long term space missions. The influences of physical, chemical, and biological parameters (e.g., liquid recirculation rate, air flow rate, support matrix surface area, matrix depth, Henry's law constant) on the removal efficiency for various gaseous contaminants were studied. The simulation results suggest that biodegradation of the trace contaminants by biofiltration is most significantly influenced by Henry's law coefficients. High removal efficiencies can be obtained for compounds with relatively low values of Henry's Law coefficient (e.g., butanol, acetone). Pollutants with high Henry's law coefficients (e.g., methane, ethylene, carbon monoxide) are difficult to eliminate in a biofilter. This can be explained by the fact that these pollutants have an unfavorable gas-liquid partition, and the pollutant concentration in the biofilm is too low to sustain a high biodegradation rate. Microbial parameters such as half-saturation coefficients also influence removal efficiency. Removal efficiencies for different compounds as a function of bioreactor packed bed depth were also predicted. Modeling efforts are also directed towards taking into consideration the effects of reduced gravity on the biofiltration process. The water film thickness is predicted to increase at reduced gravity, which might be disadvantageous for biofilter operation, due to mass transfer resistance of contaminants and oxygen. The one-box model coupled with the biofiltration model was used to predict cabin air quality and biofiltration performance for the purposes of guiding the design of ALS biofilters. Using the one box model and assuming complete removal of all major trace contaminants by the biofiltration process, the contaminant concentrations present in the cabin for 600-d travel with 6 crew members were predicted at different air flow recirculation rates Completion of BREATHe II Lab Setup. Over the past 9 months, we have completed the setup of a state-of-the-art lab for testing the BREATHe II biofiltration system. The ersatz cabin air stream is generated using permeation ovens for acetone and n-butanol, and double pattern needle valves for injection of carbon monoxide, methane, ethylene, and ammonia from concentrated gas cylinders. The ersatz air stream was prehumidified to a desired humidity level prior to gas injection by double pattern needle valve control of steam generated by a steam generator. We use computer controlled solenoid valves on the concentrated gas cylinders and a pressure switch on the gas mixing manifold to uphold lab safety and prevent shock loading of the bioreactors in the event of power failure or air compressor failure. Using precision orifices, a stainless steel air supply manifold uniformly distributes the ersatz cabin air to each bioreactor. Liquid recirculation systems for the biotrickling filters and the liquid spray systems for the biofilters were designed and constructed to distribute liquid over the reactor packing material. The Fourier Transform Infrared (FTIR) spectrometer gas analyzer conducts multi-component analysis of methane, CO2, n-butanol, ammonia, acetone, ethylene, and carbon monoxide of bioreactor inlets and outlets. The laboratory is capable of testing up to 24 bioreactors with identical inlet concentrations and flow rates. Testing multiple bioreactors simultaneously facilitates powerful factorial experimental design by operating them at different reactor configurations, packing medium types, gas residence times, and liquid types. The computer-controlled sampling and measuring systems allow automatic sequential gas sampling and measurement of contaminant concentrations, airflow rates, air temperature and relative humidity, etc., from each of the 24 bioreactors. The data acquisition and control system, achieved using LabVIEW™ software, allowed automatic, continuous, and real-time gas monitoring and data collection. Biofiltration Experiment. The first experiment is underway to investigate the feasibility of biofilters and biotrickling filters to remove multiple contaminants most commonly found in cabin air during long duration space missions. The model waste gas stream contains a five-component mixture of acetone, n-butanol, methane, ethylene, and ammonia at influent concentrations ranging from 6 to 25 ppmv. Ammonia was not injected to the air stream initially because of technical problems, but was introduced on day 30 of the test. The prehumidified waste gas stream (90% RH) was distributed evenly to six biofilters and four biotrickling filters for biological treatment, each operated with an empty bed residence time (EBRT) of 30 s. The airflow of each reactor is 7.5 L/min, and nutrients are recirculated throughout the biotrickling filters at 0.2 L/min, while 100 mL of nutrient is sprayed onto the top of the biofilters every 12 h. The packing media types among the ten reactors include perlite (2), polyurethane foam (2), and a mixture of compost, wood chips, and straw (2) in the biofilters and perlite (2) or polyurethane foam (2) in the biotrickling filters. The bioreactors were inoculated with activated sludge from a local wastewater treatment plant except the compost biofilters. Duplicate reactors were run for each of the five operating strategies. Acetone and n-butanol were degraded completely within 4 to 10 days after reactor startup for all the reactors except the foam biofilters. Lower acetone and n-butanol removal efficiencies observed in foam biofilters were consistent with the visual observation that biomass established much more slowly on the foam biofilter beds than other reactors. Methane and ethylene removal efficiencies remained extremely low within 25 d of reactor startup for all reactors, indicating the slow acclimation and proliferation of methane- and

ethylene-degrading microorganisms. The significant differences observed between the removal efficiencies of acetone, n-butanol, methane, and ethylene were consistent with conclusions of computer simulations, i.e., pollutants with high Henry's Law coefficients (e.g., methane, ethylene) are difficult to eliminate with biofiltration. Since the experiment is still in the startup phase, the system may need more time to establish the microbial consortia capable of degrading each pollutant, and long-term continuous operation of the bioreactors is therefore warranted to assess the feasibility of biofiltration for treating such multi-component trace contaminants. The lessons learned and the results of the current test will be used to design the next experiment.

For LiFT: Theoretical Modeling For successful design and operation of freeze concentration processes, the determination of the temperature bounds on water recovery is required in addition to the identification of the limiting components. The complexity arises due to the number of salts, salt hydrates and other complexes, acids, and bases in the solution and their interactions with each other. Each of these components will form a eutectic with water. Theoretical modeling, or solution phase diagram development, consists of the equilibrium observations between multi-component mixtures or solutions as a function of composition and temperature. Another point to consider are that complex mixtures consisting of several components, such as the ersatz urine solution, can be reasonably condensed to binary and/or ternary models despite the fact that solute addition progressively depresses the freezing point of solution. Multi-component (greater than two components) solutions have proven to be difficult without the aid of computer modeling, which in itself is an expensive proposition and could have a relatively low level of precision since computer models are based on chemical literature data and thermodynamic properties of solution that have never been experimentally proven. In essence, these computer models are utilized as a general process operation predictor to narrow down the range of operating temperatures required for the freeze concentration process. Thus, binary, and in some cases ternary, models can be developed based on experimental and literature data that should provide reasonable estimates of operating parameters that are not significantly different from those generated by computer models. The theoretical modeling approach was based on the binary system of urea / water since it has been determined that urea is most probably the major limiting component for urine freeze concentration. Therefore, a theoretical freezing point depression curve was determined based on the solute/solvent composition as well as the molal freezing point depression constant of the solvent, water. The resulting curve can be fitted to a second-order polynomial equation and then compared to experimental data. The results of the eutectic point determination experiment indicate a very good correlation between predicted and actual results. An initial examination of the results reveals a step decrease in temperature between the theoretical freezing point depression curve and the experimental results for the urea extraction due to supersaturation conditions of urea in the solution. Inorganic / Organic Migration This multi-faceted objective was to determine contaminant migration into the product water for both a ternary urine solution (urea, sodium chloride, and water) and the transit ersatz urine wastewater solution. From the results for the ternary solution it is obvious that the freeze concentration process is potentially a feasible option for potable water production from a urine waste stream; however, to facilitate a better conception of the capability of the freeze concentration process, the multi-component ersatz solution was tested. This portion of the study was based on the results of the ternary solution with the purpose of determining appropriate variables in the operation. Antibiotic and Biological Removal Since the beginning of this research emerging concerns have developed in regards to other contaminants that may enter the water cycle of an advanced life support system. These concerns correlate with the recent interest in the environmental field that has focused attention on low level contamination issues of personal care products and pharmaceuticals in water bodies used as a source of potable water. Space based resource recovery can mimic Earth bound environmental contamination problems especially in the case of antibiotic contamination. In the event of crew member illness, a large portion of non-metabolized antibiotic will be released to the water cycle through the urine; therefore, developing a technology that is capable of removing this contaminant from the waste stream is required to prevent potential negative downstream reuse impacts such as unintended human consumption and the development of bacterial resistance. The main objective of this study was to determine the capability of the freeze concentration process in the removal of Tylosin. Initial estimates predicted high removal rates due to the fact that Tylosin, along with many other antibiotics, is a relatively large molecule. As predicted the freeze concentration process was successful in removing a vast majority of the Tylosin, below the minimum detectable limit, in the model urine solution thus making it a good candidate for low level contaminant removal in a closed loop system. Antibiotic and Biological Removal Since the beginning of this research emerging concerns have developed in regards to other contaminants that may enter the water cycle of an advanced life support system. These concerns correlate with the recent interest in the environmental field that has focused attention on low level contamination issues of personal care products and pharmaceuticals in water bodies used as a source of potable water. Space based resource recovery can mimic Earth bound environmental contamination problems especially in the case of antibiotic contamination. In the event of crew member illness, a large portion of non-metabolized antibiotic will be released to the water cycle through the urine; therefore, developing a technology that is capable of removing this contaminant from the waste stream is required to prevent potential negative downstream reuse impacts such as unintended human consumption and the development of bacterial resistance. In case of the possibility of crew member urinary tract infections, the freeze concentration process may prevent biological migration to the product water; therefore, a biological migration study was performed. Despite the fact that urine is normally a sterile waste stream it may contain human pathogens, most predominantly E. coli, such as the case in a urinary tract infection. Therefore, the possibility of microbial migration from the urine solution to the product water was studied on the batch freeze concentration process by doping the ersatz urine solution utilizing the ATCC 25922 strain of E. coli. Each control solution, distilled water, the ersatz urine solution, and tryptic soy broth was doped and underwent three separate scenarios to compare the vitality of the E. coli in each case. Three cases were considered. The first was to determine the number of colony forming units per 100 mL in an initial control solution. It can be assumed that the initial values between the different control groups cannot be compared since the doped count of E. coli may vary between freeze-dried ampules, and dilutions thereof. The latter cases set the baseline for E. coli vitality for ambient conditions and for a basic freeze-thaw cycle condition. Once the control was defined the ersatz solution was doped and underwent the freeze concentration process to determine the migration of E. coli into the product water stream. From these results it is obvious that the freeze concentration process is effective in removing greater than 99% of the E. coli in the influent stream. Further tests would need to be performed in order to verify this result on a continuous freeze concentration process. Urea Recovery This exciting research effort will focus on the potential recovery of urea along with trace organics to be utilized as a nutrient source for plants in a long-term advanced life support initiative. The foundation of this recovery lies in the fact that the eutectic point of urea lies above the eutectic point of NaCl therefore its separation from salt brine may be possible. In order to achieve such separation a dual freeze concentration system aligned in series may be the optimal method. The first concentration would extract water while the second concentration would extract a urea / water solution. Future experiments based on the utectic theoretical model will pursue the possibility of this r ecovery.

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